Advanced Data Center Energy Opportunities: Cloud and Infrastructure CoP—Data Center Energy and Efficiency With AI Adoption

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Data Center Energy Demand

A recent Lawrence Berkeley National Laboratory (LBNL) study, incorporating changes observed in the data center sector in recent years, indicates "that the electricity consumption of U.S. data centers is currently growing at an accelerating rate" (Shehabi et al. 2024).

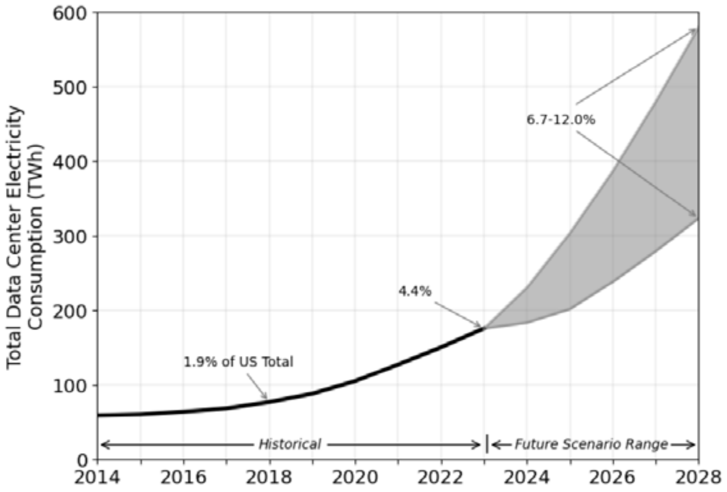
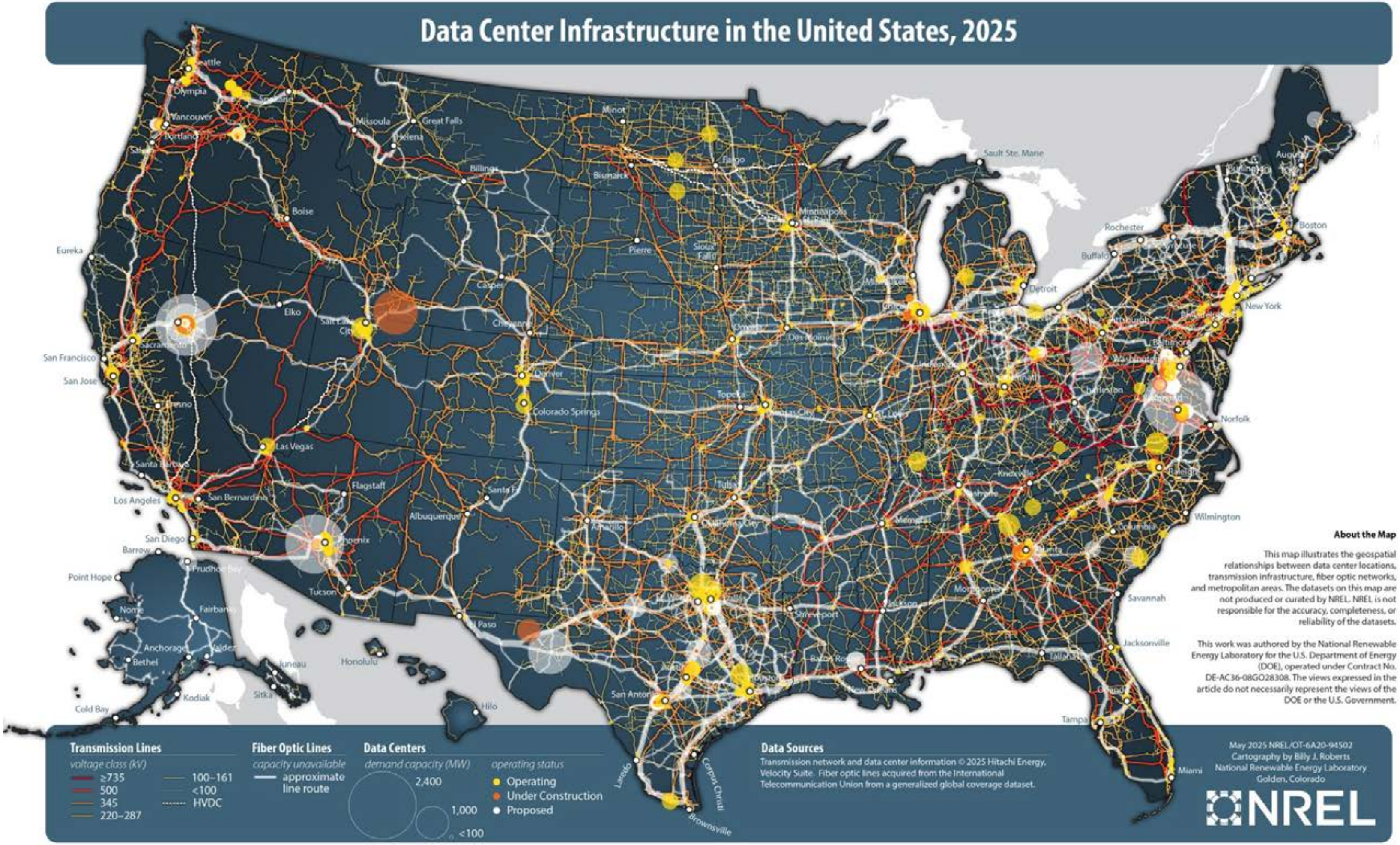


Figure ES-1. Total U.S. data center electricity use from 2014 through 2028.

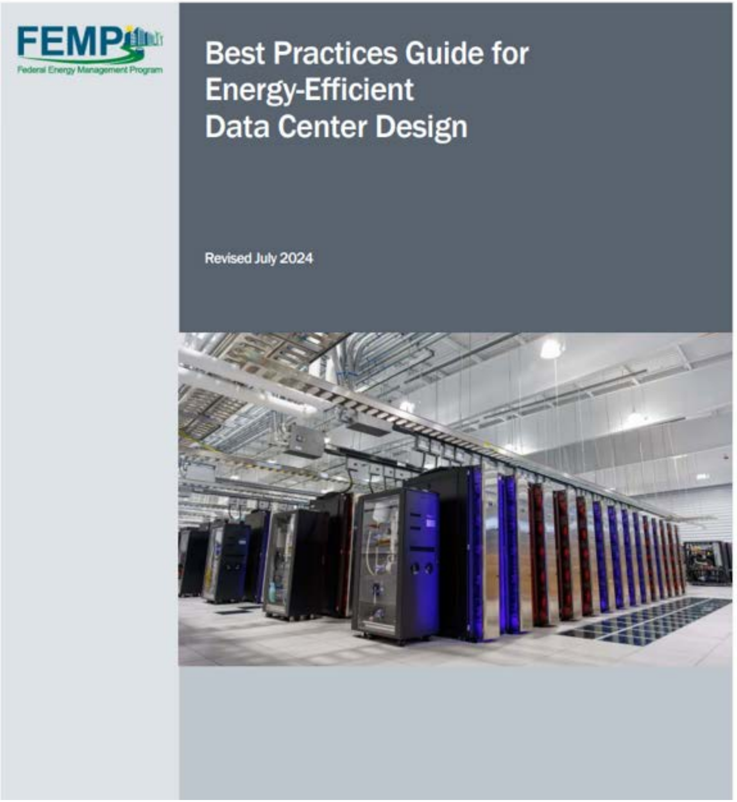
Figure ES-1 from the LBNL data center energy use report shows a **compound annual growth rate of approximately 13% to 27% between 2023 and 2028 (from approximately 200 TWh to a range of 325 TWh to 580 TWh**. Note: This surge in demand should be understood within the context of a much larger electricity demand expected from other factors (e.g., increase in electric vehicle adoption). The report provides details on energy use by data center type and industry trends.

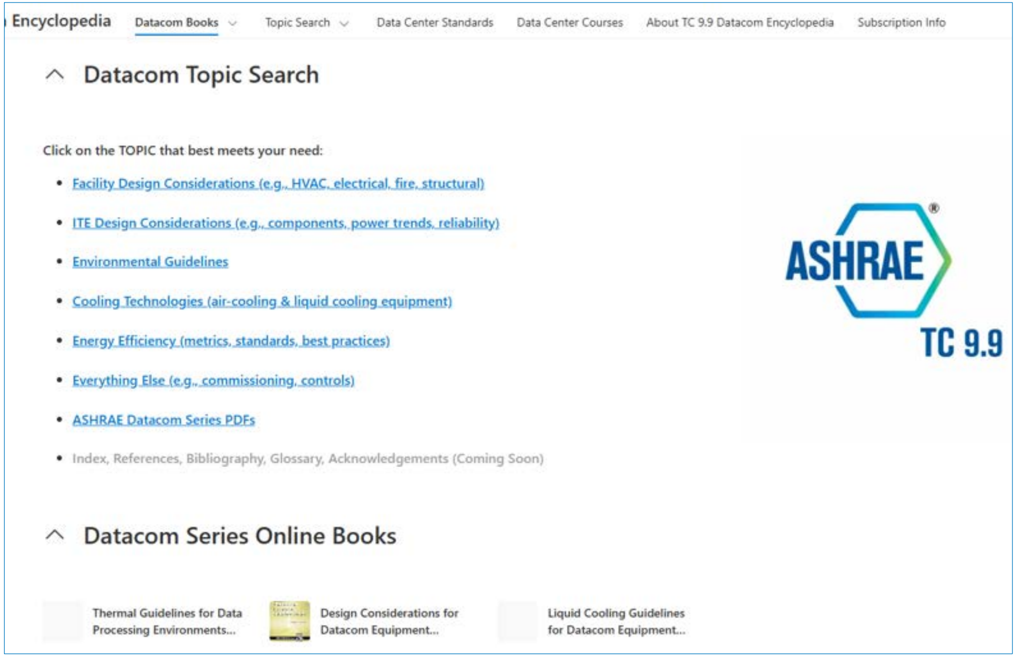
Source: Shehabi, A., S. J. Smith, A. Hubbard, A. Newkirk, N. Lei, M. A. B. Siddik, B. Holecek, J. Koomey, E. Masanet, and D. Sartor. 2024. *2024 United States Data Center Energy Usage Report*. Berkeley, CA: Lawrence Berkeley National Laboratory. LBNL-2001637. doi.org/10.71468/P1WC7Q.



Source: Roberts, B. 2025. Data Center Infrastructure in the United States, 2025 (map). docs.nrel.gov/docs/gen/fy25/94502.jpg.

Follow Recommendations in *Best Practices Guide for Energy-Efficient Data Center Design*, ASHRAE

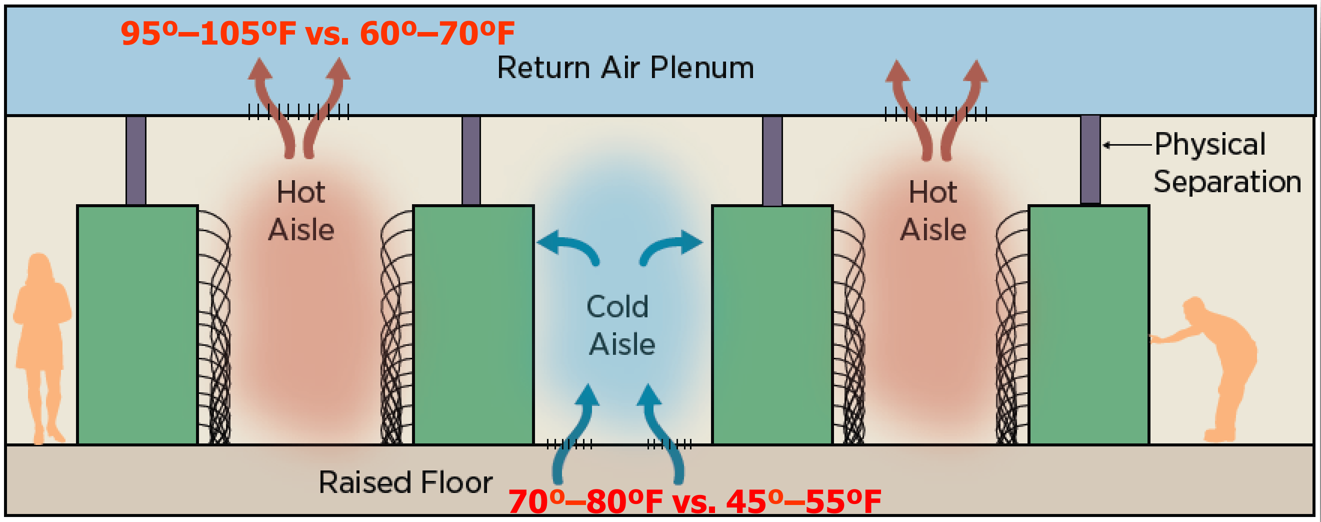




Source: ASHRAE. 2025. "Data Center Resource Page." https://www.ashrae.org/technical-resources/bookstore/datacom-series.
(TC 9.9 Datacom Encyclopedia subscription includes access to PDFs of all prior TC 9.9 books.)

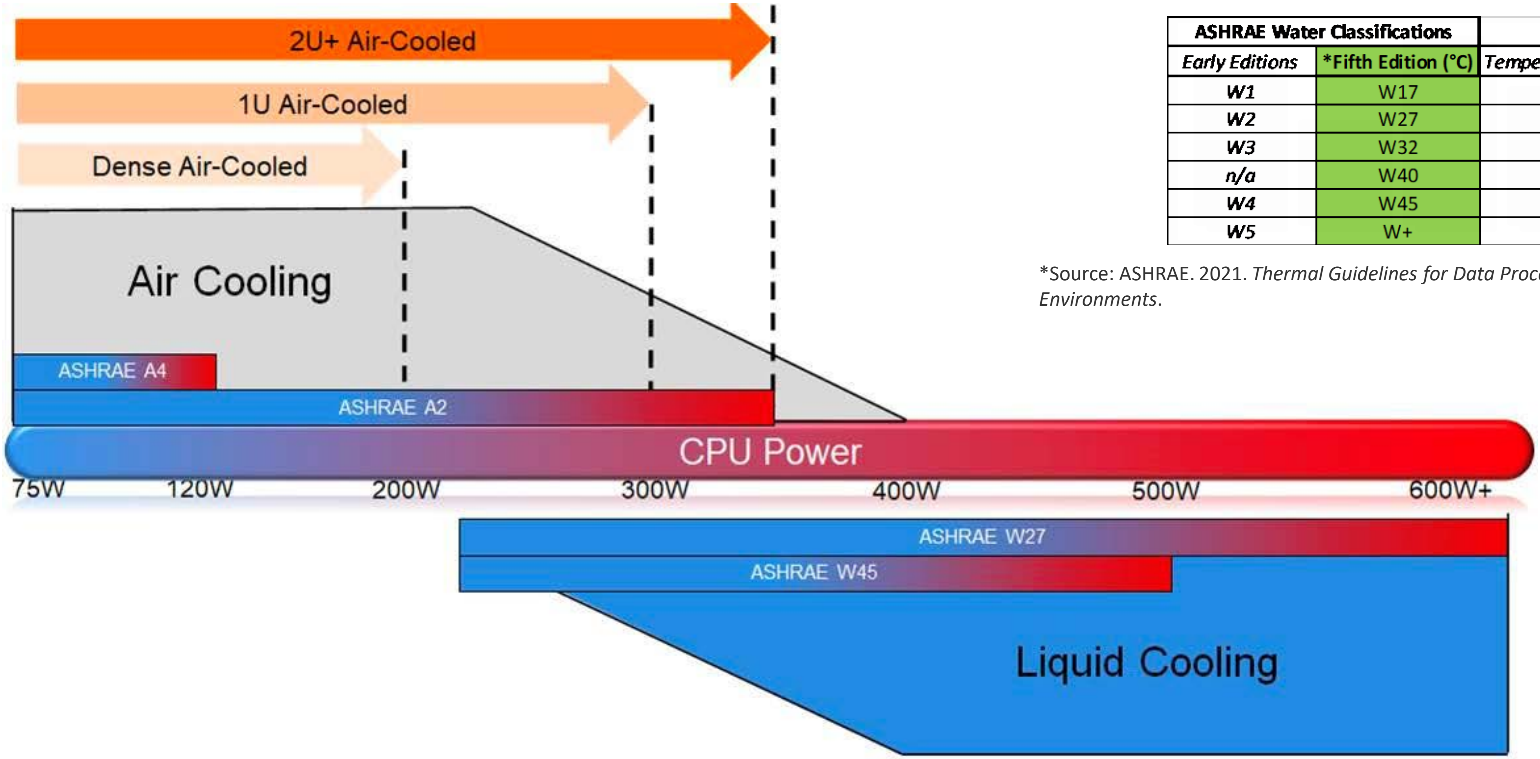
Source: DOE. 2025. "Best Practices Guide for Energy-Efficient Data Center Design." FEMP. www.energy.gov/femp/articles/best-practices-guide-energy-efficient-data-center-design

Isolate Cold and Hot Aisles



Source: www.energy.gov/sites/prod/files/2013/10/f3/eedatacenterbestpractices.pdf

Increasing CPU Power
Transition to Warm Water Cooling



ASHRAE Water Classifications		
Early Editions	*Fifth Edition (°C)	*Temperature (°F)*
W1	W17	*62.6*
W2	W27	*80.6*
W3	W32	*89.6*
n/a	W40	*104.0*
W4	W45	*113.0*
W5	W+	*> 113.0*

*Source: ASHRAE. 2021. *Thermal Guidelines for Data Processing Environments*.

Air cooling versus liquid cooling, transitions, and temperatures.

Source: Figure 4 from ASHRAE 2021 white paper “Emergence and Expansion of Liquid Cooling in Mainstream Data Centers,” tpc.ashrae.org/Documents?cmtKey=fd4a4ee6-96a3-4f61-8b85-43418dfa988d.

Data Center Metrics

Data Center Metrics: The Green Grid PUE "Family"

Energy:

$$PUE = \frac{\text{"}\textit{Facility energy}\text{"} + \text{"}\textit{IT energy}\text{"}}{\text{"}\textit{IT energy}\text{"}} = \frac{\text{"}\textit{Total energy}\text{"}}{\text{"}\textit{IT energy}\text{"}}$$

$$ITUE = \frac{\text{"}\textit{IT energy (includes server fans, CDU, power conversion)}\text{"}}{\textit{Total energy into the }\text{"}\textbf{Compute components}\ (\textit{CPU, memory, storage})\text{"}}$$

$$TUE = PUE \ x \ ITUE$$

$$ERE = \frac{\text{"}\textit{Facility energy}\text{"} + \text{"}\textit{IT energy}\text{"} - \text{"}\textbf{Reuse energy}\text{"}}{\text{"}\textit{IT energy}\text{"}}$$

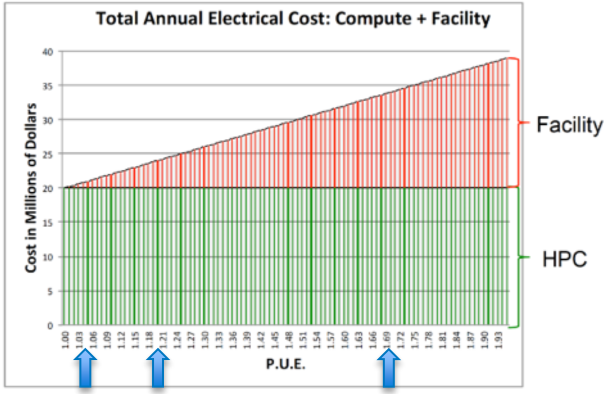
Sustainability (*water*):

$$WUE = \frac{\text{"}\textit{Annual site water usage}\text{"}}{\text{"}\textit{IT energy}\text{"}} \qquad \left(units \ \frac{L}{kWh}\right)$$

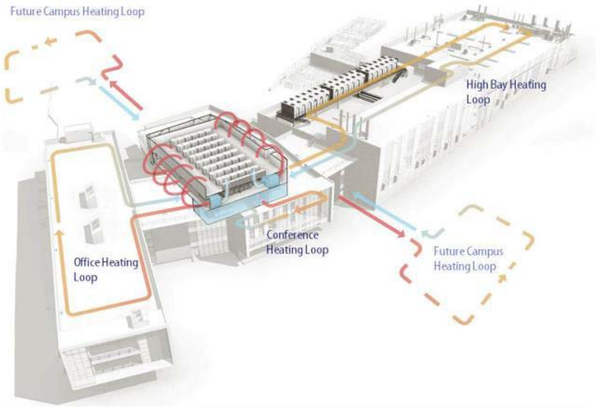
$$WUE_{source} = \frac{\text{"}\textit{Annual site water usage}\text{"} + \text{"}\textit{Annual source energy water usage}\text{"}}{\text{"}\textit{IT energy}\text{"}}$$

Utilization (*based on industry*):

$$\{Gflops,\ Transaction\}\ per\ watt$$

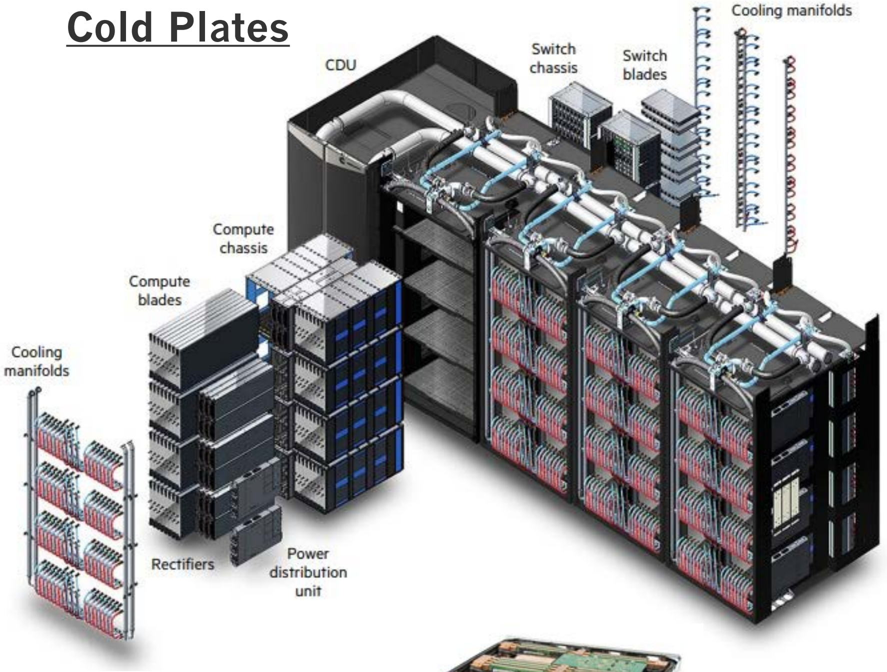


Assume ~20MW HPC system & $1M per MW year utility cost.



Fanless Direct-Liquid-Cooled Server Options

Cold Plates



Images from HPE

Data Center Metrics: PUE "Family"

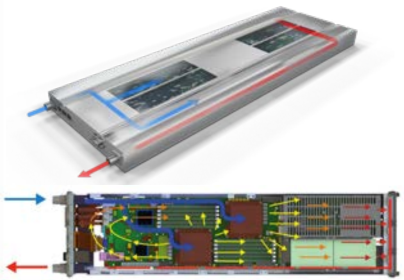
Energy

$$PUE = \frac{\text{"Facility energy"} + \text{"IT energy"}}{\text{"IT energy"}} = \frac{\text{"Total energy"}}{\text{"IT energy"}}$$

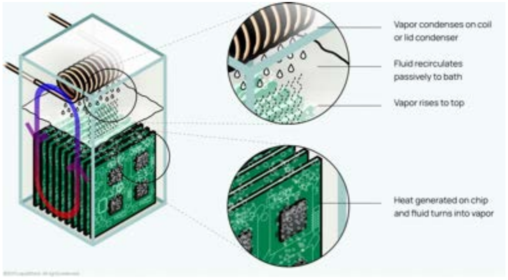
$$ITUE = \frac{\text{"IT energy"}}{\text{Total energy into the "Compute components"}}$$

$$TUE = PUE \text{ x } ITUE$$

Direct Immersion

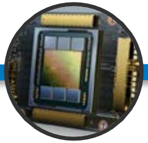


Single Phase



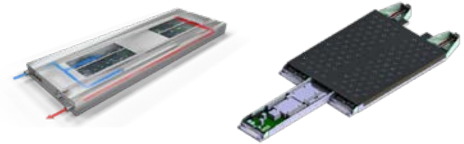
Two Phase

Image from Liquid Stack

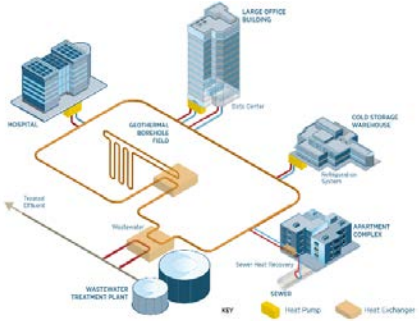



Advanced Liquid Cooling Enabling Heat Reuse

NREL has been directly involved in a number of independent tests of new cold plates, immersion and was awarded the testing and evaluation funding for the ARPA-E COOLERCHIPS program.



- **CHIP-level thermal resistance:** ACO, Georgia Tech, and APEEM group collaborating to measure thermal resistance at the chip level
- **Server-level efficiency measurement:** Developing testing protocols for measuring server-level efficiency, providing guidance to measure iTUE
- **Data center-scale effects based on server measurement:** Scale up from detailed server testing to macro effects on data center efficiency.



Production Liquid Cooling

Three generations of HPC systems: Generation 1 was prototype hardware, generation 2 added hybrid cooling, and generation 3 extreme high density all DLC.

Decade of lessons learned:

- Production liquid cooling
- Recommissioning
- Live upgrades.

Data Center Modeling

Developing integrated co-simulation software: Optimize design of data centers including their data, power, and thermal management systems for lowering cooling energy demand, lowering CO2 footprint, and lowering cost while maintaining reliability and availability.

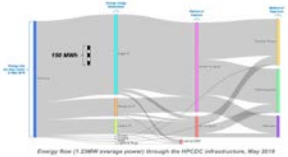
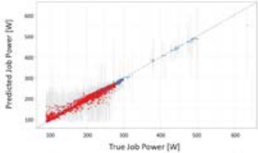
To facilitate the adoption of the tool to achieve transformational and disruptive design advances first for COOLERCHIPS and then by the larger data center design community.

Modular Data Center Design

Expanded from 2.5, 5 and 7.5 MW

Upgrades while systems are in production. Controls augmentation, lessons learned

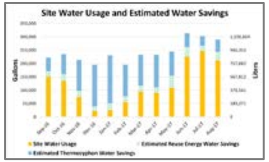
Energy Measurements

Measuring computing energy/carbon intensity for user visibility, forecasting, and system cross-comparison:

- **Workload profiling** across different application types
- **Accounting standardization** for carbon intensity accounting.

Water Usage



- **Water usage:** Direct reduction in water usage through application of advanced dry coolers (Sandia National Labs and Johnson Controls)
- **Heat reuse:** Comparing and contrasting methods for carbon intensity accounting across industry and public sector.

NREL’s Key Approach to Optimize Data Centers

1. Power usage effectiveness (PUE): Reduce energy use by making systems as efficient as possible:
   - ***Maximize compute <u>entering</u> temperature to maximize <u>energy efficiency</u>!***
   - “Free” cooling to reduce or eliminate compressor (chiller, DX)-based cooling:
     a) Install direct liquid-cooled computers that use warm water.
     b) Capture as much heat as possible directly to the liquid cooling system.
   - Optimize fan/pump speeds and uninterruptible power supply.
2. Energy reuse effectiveness (ERE): Heat reuse—achieve as low an ERE as possible:
   - ***Maximize compute <u>leaving</u> temperature to maximize <u>energy reuse</u>!***
3. Water usage effectiveness (WUE): Reject as much remaining heat to dry coolers as possible:
   - ***Maximize compute <u>leaving</u> temperature to maximize <u>heat-rejected dry</u>!***

Questions?

www.nrel.gov

NLR/PR-6A40-96792

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